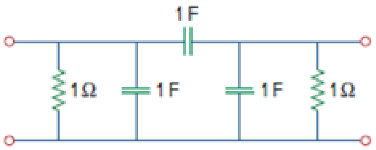
Chapter 19, Problem 49P

Problem

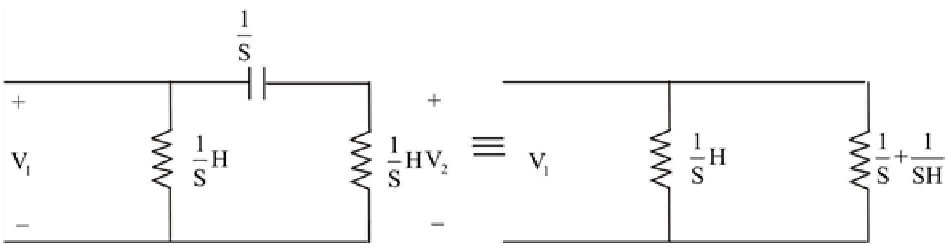
Using impedances in the *s*-domain, obtain the transmission parameters for the circuit in Fig. 19.103.

Figure 19.103



Step-by-step solution

Step 1 of 7



Step 2 of 7

$$I_2 = \frac{I_1 \times \left(\frac{1}{S}+1\right)}{\frac{1}{S+1}+\frac{1}{S}+\frac{1}{S}+1}$$

$$= \frac{I_1\left(\frac{1}{S+1}\right)}{\frac{2S+S+1}{(S+1)S}} = \frac{SI_1}{3S+1}$$

$$V_1 = I_1\left(\frac{1}{S+1} \| \left(\frac{1}{S}+\frac{1}{S+1}\right)\right)$$

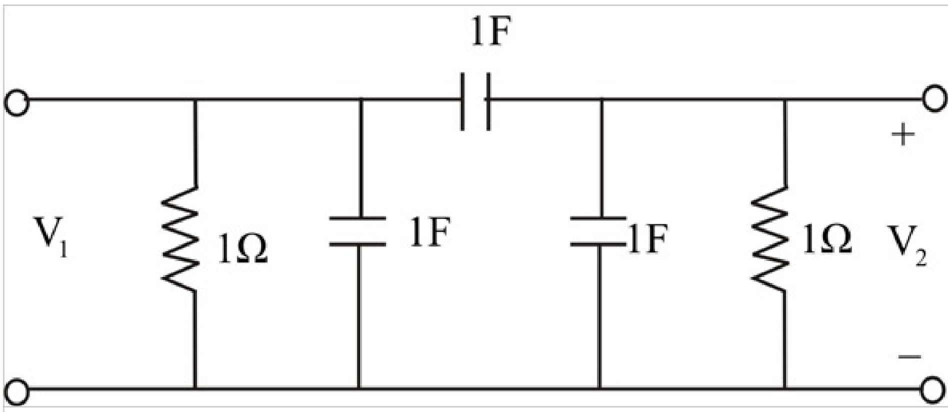
$$V_1 = I_1\left(\frac{1}{S+1} \| \frac{2S+1}{S(S+1)}\right)$$

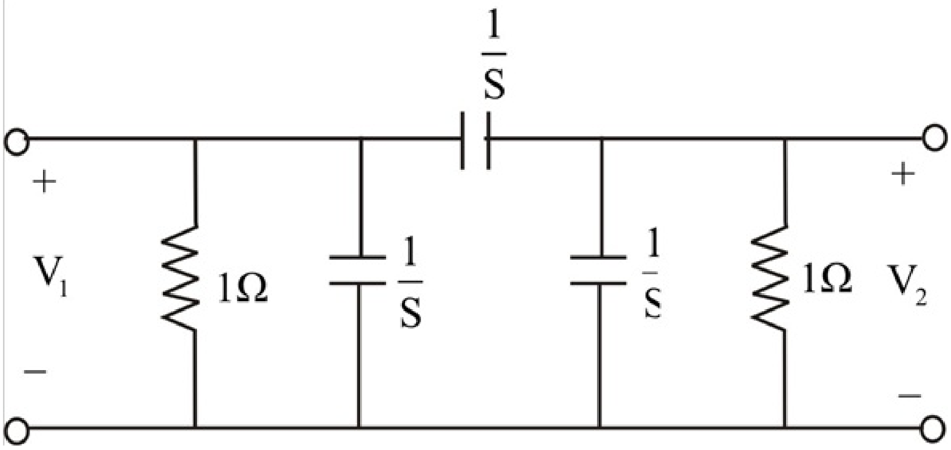
$$= I_1\left(\frac{\frac{2S+1}{S(S+1)}}{\frac{3S+1}{S(S+1)}}\right) = \frac{I_1(2S+1)}{(3S+1)(S+1)}$$

Step 3 of 7

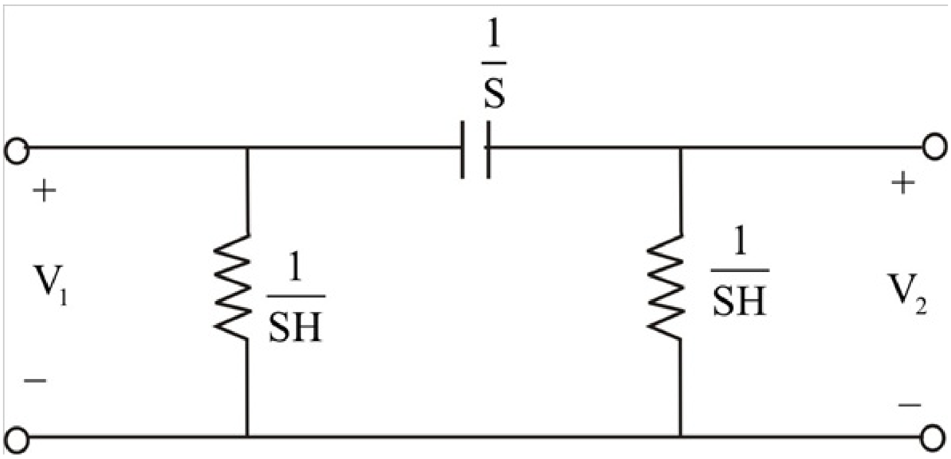
$$\frac{V_1}{V_2} = \frac{I_1(2S+1)}{(3S+1)(S+1)} \times \frac{(3S+1)\times(S+1)}{SI_1} \cdot \frac{2S+1}{(S+1)}(S+1)$$

$$A = \frac{2S+1}{S(S+1)} \times (S+1).2 + \frac{1}{S}$$





$$\left(1 \| \frac{1}{S}\right) = \frac{\frac{1}{S}}{1+\frac{1}{S}} = \frac{1}{S+1}$$



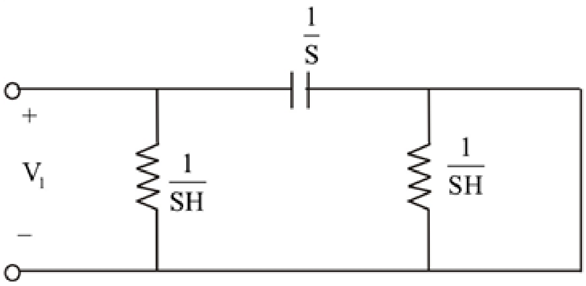
$$A=\left.\frac{V_1}{V_2}\right|_{I_2=0} \qquad B=\left.\frac{-V_1}{I_2}\right|_{V_2=0} \qquad C=\left.\frac{I_1}{V_2}\right|_{I_2=0} \qquad D=\left.\frac{V_1}{I_2}\right|_{V_2=0}$$

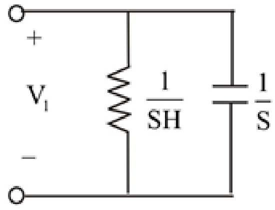
Setting $I_2=0$

$$\frac{V_1}{V_2}=\frac{2S+1(+S1)}{S(S+1)}$$

$$\Rightarrow \frac{I_1(2S+1)}{(3S+1)(S+1)V_2}=\frac{2S+1}{S(S+1)}$$

$$\frac{I_1}{V_2}=\frac{(3S+1)(S+1)}{S}$$





Step 7 of 7

$$B=\frac{-V_2}{I_2}=\frac{+I_1\left(\frac{1}{S}+1\right)}{\frac{1}{S}+1+\frac{1}{S}}=I_2$$

$$=\frac{+I_1\left(\frac{1}{S}\times\frac{1}{S+1}\right)}{\left(\frac{1}{S}+\frac{1}{S+1}\right)}\times\frac{\left(\frac{1}{S}+\frac{1}{S}+1\right)}{I_1\left(\frac{1}{S}+1\right)}=\frac{1}{S}$$

$$D=\frac{-I_1}{I_2}$$

$$=\frac{-I_1\left(\frac{1}{S+1}+\frac{1}{S}\right)}{+I_1\left(\frac{1}{S}+1\right)}=\frac{(2S+1)}{S}=2+\frac{1}{S}$$

$$=\begin{bmatrix} 2+\frac{1}{S} & \frac{1}{S} \\ \frac{(3S+1)(S+1)}{S} & 2+\frac{1}{S} \end{bmatrix}$$